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BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Mechanical

ROBOTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. First part of each question is compulsory and attempt any two from the rest. Give neat and clean diagrams wherever needed.

Q. 1. (a) Define a robot. 2

(b) Briefly describe three basic configurations of arm in robotic manipulators. 7

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(c) What are design and control issues in robotics ? Explain briefly. 7

(d) Write short note on Robot Anatomy and notations used in robotics. 7

Q. 2. (a) What is mapping. 2

(b) A vector $p = 3\hat{i} - 2\hat{j} + 5\hat{k}$ is first rotated by 60° about x-axis, then by 45° about z-axis. Finally, it is translated by $-3\hat{i} + 2\hat{j} - 5\hat{k}$. Determine the new position of vector p. 7

(c) Derive the fundamental rotation matrices about x, y and z axis from rotational transformation matrix. 7

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(d) Frame {2} is rotated with respect to frame {1} about the x-axis by an angle of 45°. The position of origin of frame {2} as seen from frame {1} is ${}^{1}D_2$ = [6.0 5.0 4.0] obtain transformation matrix ${}^{1}T_2$, which describes frame {2} relative to frame {1}. Also, find the description of point p in frame {1} if ${}^{2}p$ = [4.0 6.0 8.0]. 7

Q. 3. (a) What is forward & inverse kinematics ? 2

(b) Explain Denavit Hartenberg notations with the help of diagram. 7

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(c) Determine the overall transformation matrix for the 3 DOF (RPP) manipulator arm shown in figure 1. 7

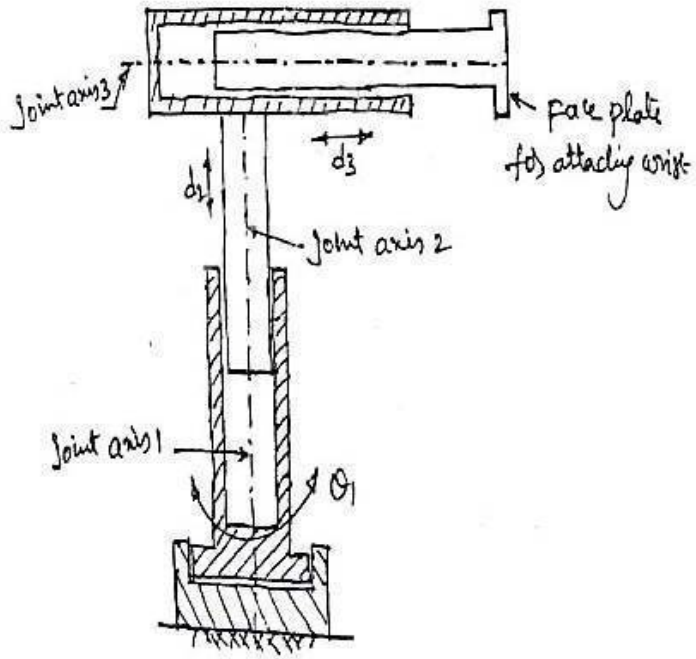

Figure 1 : 3 DOF cylindrical (RPP) manipulator arm

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(d) Obtain the position and orientation of the tool point p with respect to the base for the 2 DOF, RP planar manipulator as shown in figure 2. 7

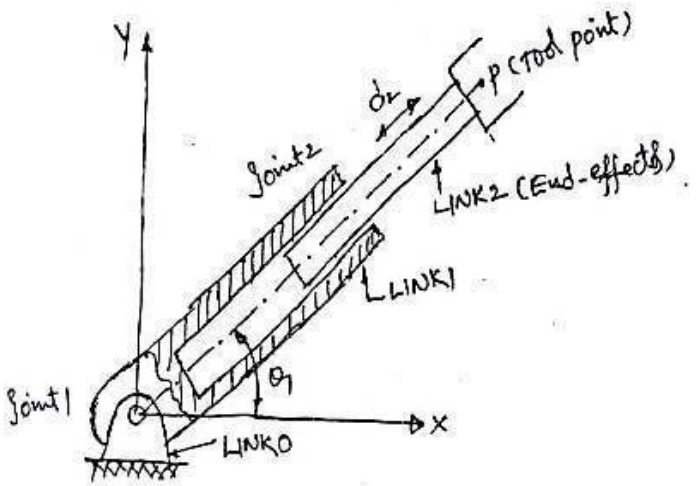

Figure 2 : A 2-DOF planar manipulator arm

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Q. 4. (a) What are functions of sensor in robotics. 2

(b) Explain various kinds of sensors used in robotics with suitable diagram. 7

(c) What are the industrial applications of vision controlled robotic systems. 7

(d) Briefly discuss a robotic vision system with multiple cameras as for manipulator control. 7

Q. 5. (a) What are the Non-Industrial applications (mention any four). 2

(b) What are various material handling tasks performed by a robot. 7

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(c) Explain in detail industrial applications of robot with suitable diagram. 7

(d) Write short notes on any two : 7

(i) Justifications of robots

(ii) Robot safety

(iii) Inspection applications

4

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